

Lorentzian Boundary Kernels as a Compactification Test for Primordial Cosmology

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A proposed higher-dimensional universe is usually tested against the cosmic microwave background only after a complete four-dimensional effective theory has been constructed, canonically normalized, and evolved mode by mode. We formulate a complementary test in which the primary object is the constrained Lorentzian boundary kernel of the Feynman–DeWitt wave functional. At quadratic order the imaginary part of this kernel is the inverse covariance of the physical boundary perturbations. The scalar and tensor power spectra, their tilts, the tensor-to-scalar ratio, isocurvature power, and stability conditions can therefore be read from one matrix-valued object. Heavy moduli and Kaluza–Klein modes enter through a Schur complement, which makes their decoupling or observational imprint explicit before a full Boltzmann analysis is attempted. We derive the general construction, recover the standard inflationary result in the single-field limit, and give two worked compactification templates: $M_4 \times S^2 \times \mathbb{CP}^2$, including the flux branches $(m, n) = (3, 3)$ and $(1, 5)$, and $M_4 \times S^3 \times S^3$ as a geometrically distinct control. The method does not uniquely reconstruct an internal manifold from the CMB. It provides a rapid acceptability test and a discriminant whenever internal modes remain light enough to modify the scalar or tensor kernel.

I. INTRODUCTION

The usual route from a higher-dimensional theory to a prediction for the cosmic microwave background (CMB) is conceptually straightforward but technically expensive. One chooses a compactification, reduces the theory to four dimensions, identifies the light fields, transforms to the four-dimensional Einstein frame, diagonalizes the kinetic terms, finds an inflationary background, derives the gauge-invariant perturbation equations, selects a quantum state, evolves every relevant mode, and finally passes the primordial spectra to a Boltzmann code. This sequence is necessary for precision parameter estimation. It is often more work than is needed to decide whether a candidate universe is viable at all.

This paper develops a faster intermediate test. The central object is the quadratic boundary kernel of the Lorentzian Feynman–DeWitt wave functional. The kernel is evaluated on the classical saddle after the lapse, shift, gauge redundancy, and nondynamical matter variables have been eliminated. Its imaginary part determines the Gaussian width of the physical boundary state. In Fourier space that width is the primordial covariance. In a coupled system the kernel is a matrix containing the curvature perturbation, the two graviton polarizations, isocurvature fields, moduli, and any Kaluza–Klein (KK) modes retained in the truncation.

The method is not a replacement for standard cosmological perturbation theory. At quadratic order it is the same physics written in a form particularly well adapted to comparing compactifications. The advantage is organizational. All effects of the internal manifold enter through four ingredients: the background saddle, the kinetic metric of the light fields, the mass and mixing matrices, and

the spectrum of internal differential operators. These ingredients flow into a single effective boundary kernel. A compactification can be quickly shown to be consistent with present day observations of the CMB, or it can fail quickly because the kernel has a negative-width direction, an unsuppressed isocurvature eigenmode, an unacceptable scalar tilt, an excessive tensor amplitude, or a light internal mode that produces large features.

The formulation adopted here is therefore not proposed as an alternative quantum theory. For a fixed background sector, contour, and Gaussian state, it is equivalent to the Schrödinger wave functional, canonical mode-function quantization, and the in-in formalism. Its value lies in the way it organizes the problem. The classical solution appears as a saddle of the action, while the quadratic boundary Hessian records the finite quantum width around that saddle. Compactification moduli, KK modes, and observable perturbations can consequently be assembled into one matrix kernel and reduced uniformly before comparison with cosmological data. This representation may become especially useful when several fields, compactification sectors, or saddles must be retained simultaneously.

The uncertainty principle and inflationary mode evolution are likewise not treated as two independent stochastic mechanisms. In the Gaussian approximation they are, respectively, the width of the quantum state and the subsequent dynamical evolution of that same width. Inflation does not generate statistically independent fluctuations point by point; it evolves a correlated field configuration whose covariance is encoded by the boundary kernel. This viewpoint changes no quadratic prediction, but it makes explicit which information belongs to the state, which belongs to propagation, and which assumptions enter when a particular compactification and contour are selected.

The relation between a Gaussian wave functional and a cosmological spectrum is standard [1–3]. The useful point here is to treat the constrained matrix kernel as a common interface between higher-dimensional geometry

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and CMB observables. The same formula applies when one changes the compactification manifold, flux sector, or modulus content. In the decoupling limit all acceptable compactifications reduce to the familiar four-dimensional result. A manifold can be distinguished only through corrections that survive in the observable window.

We illustrate the procedure with two ten-dimensional products. The first is

$$\mathcal{M}_{10} = M_4 \times S^2 \times \mathbb{CP}^2, \quad (1)$$

with independent radii and quantized fluxes. It admits the factorized chiral index

$$N_{\text{fam}} = m \frac{n^2 - 1}{8}, \quad (2)$$

so that both $(m, n) = (3, 3)$ and $(1, 5)$ yield three chiral families while distributing the flux and zero-mode wave functions differently over the two factors. The second example is

$$\mathcal{M}_{10} = M_4 \times S^3 \times S^3, \quad (3)$$

which has a different curvature, harmonic spectrum, and modulus structure but remains simple enough to display the kernel construction analytically. It serves as a control rather than as a complete particle-physics model.

II. THE CONSTRAINED LORENTZIAN BOUNDARY KERNEL

A. Wave functional and saddle width

Let $q_A(\mathbf{x})$ denote a complete set of physical boundary variables on a spacelike hypersurface Σ . Before reduction, these variables are drawn from the induced spatial metric, matter fields, and internal metric components. The Feynman–DeWitt wave functional is

$$\Psi[q] = \int_{Q|\Sigma=q} \mathcal{D}Q \exp\left(\frac{i}{\hbar} S[Q]\right), \quad (4)$$

where Q collectively denotes bulk geometries and matter histories. The boundary data q are not the bulk integration variables. They label the state obtained after the bulk histories have been summed.

Suppose the integral is dominated by a saddle $\bar{Q}$ and write $Q = \bar{Q} + \eta$. After gauge fixing and elimination of constraints, the action has the quadratic expansion

$$S[\bar{Q} + \eta] = S[\bar{Q}] + \frac{1}{2} \eta \cdot \mathcal{A} \cdot \eta + \dots \quad (5)$$

The contour that defines the state selects a generally complex solution of the linearized equations. Evaluating the quadratic action on that solution gives a boundary term,

$$S_{\text{os}}^{(2)}[q] = \frac{1}{2} \int \frac{d^3k}{(2\pi)^3} q_A(-\mathbf{k}) K_{AB}(k) q_B(\mathbf{k}). \quad (6)$$

The matrix K_{AB} is the Dirichlet-to-Neumann map: it converts the boundary amplitude into the canonical momentum of the regular bulk solution,

$$K_{AB}(k) = \left. \frac{\delta \Pi_A(\mathbf{k})}{\delta q_B(\mathbf{k})} \right|_{\Sigma}. \quad (7)$$

Write $K = K_R + iK_I$, with K_I Hermitian. Then

$$|\Psi[q]|^2 \propto \exp\left[-\frac{1}{\hbar} \int \frac{d^3k}{(2\pi)^3} q_A(-\mathbf{k}) (K_I)_{AB} q_B(\mathbf{k})\right]. \quad (8)$$

A normalizable Gaussian requires K_I to be positive definite on the physical subspace. The covariance is

$$\langle q_A(\mathbf{k}) q_B(\mathbf{k}') \rangle = (2\pi)^3 \delta^3(\mathbf{k} + \mathbf{k}') \frac{\hbar}{2} (K_I^{-1})_{AB}. \quad (9)$$

Equation (9) is the central diagnostic of this paper.

The interpretation is elementary. The saddle is the mean history. The quadratic neighborhood of the saddle is its quantum width. In canonical language the uncertainty relation fixes an irreducible phase-space area, while the Hessian and the contour determine the shape and orientation of the uncertainty ellipse. The path integral does not assign independent random numbers to spatial cells; it assigns an amplitude to an entire constrained field configuration. A $1/\sqrt{N}$ counting law appears only when a coarse-grained observable averages over approximately independent correlation volumes.

B. Constraints and physical variables

For gravity the ADM action has the form

$$S = \int dt d^3x \left[\pi^{ij} \dot{h}_{ij} + p_I \dot{\phi}^I - N\mathcal{H} - N^i \mathcal{H}_i \right]. \quad (10)$$

The lapse and shift are Lagrange multipliers. Their functional integration enforces the Hamiltonian and momentum constraints. At the linearized level these constraints remove scalar and vector gauge directions and relate matter and metric perturbations. A practical calculation may solve the constraints before the path integral or integrate over nondynamical variables directly; both procedures give the same reduced kernel when the measure is treated consistently.

For an isotropic four-dimensional background with scalar fields φ^a , a convenient physical basis is

$$q_A = (\zeta, s^\alpha, h_+, h_\times, \chi^r), \quad (11)$$

where ζ is the adiabatic curvature perturbation, s^α are isocurvature directions, h_λ are the two tensor polarizations, and χ^r are retained internal or KK modes. The kernel in this basis is gauge invariant. A negative eigenvalue of K_I is not a coordinate instability; it signals that the selected contour does not define a normalizable state in that direction or that the saddle is unstable.

C. Moving the boundary

The division between an “initial state” and “later mode evolution” depends on where an intermediate hypersurface is placed. Let $G_{21}(q_2, q_1)$ be the quadratic propagator between two boundaries. Composition gives

$$\Psi_2(q_2) = \int \mathcal{D}q_1 G_{21}(q_2, q_1) \Psi_1(q_1). \quad (12)$$

For Gaussian kernels this integration is exact. If

$$S_{21}^{(2)} = \frac{1}{2} q_2 A q_2 + q_2 B q_1 + \frac{1}{2} q_1 C q_1 \quad (13)$$

and the earlier state has kernel K_1 , then, up to convention-dependent factors of i ,

$$K_2 = A - B(C + K_1)^{-1} B^T. \quad (14)$$

Moving the intermediate boundary changes K_1 and the propagator separately but not the final state. One may therefore place the numerical boundary where the effective description is best controlled and evolve the kernel rather than repeatedly re-quantizing the system.

III. CMB OBSERVABLES FROM THE KERNEL

For the curvature perturbation,

$$\langle \zeta_{\mathbf{k}} \zeta_{\mathbf{k}'} \rangle = (2\pi)^3 \delta^3(\mathbf{k} + \mathbf{k}') P_\zeta(k), \quad (15)$$

$$\Delta_\zeta^2(k) = \frac{k^3}{2\pi^2} P_\zeta(k). \quad (16)$$

If all nonadiabatic modes have been integrated out,

$$P_\zeta(k) = \frac{\hbar}{2 \operatorname{Im} K_{\zeta, \text{eff}}(k)}. \quad (17)$$

The amplitude, tilt, and running follow from

$$A_s = \Delta_\zeta^2(k_*), \quad (18)$$

$$n_s - 1 = \left. \frac{d \ln \Delta_\zeta^2}{d \ln k} \right|_{k_*}, \quad (19)$$

$$\alpha_s = \left. \frac{d n_s}{d \ln k} \right|_{k_*}. \quad (20)$$

For the two graviton polarizations,

$$\Delta_T^2(k) = \frac{k^3}{2\pi^2} \sum_{\lambda=+, \times} P_{h_\lambda}(k), \quad (21)$$

where each covariance is obtained from the inverse tensor block of K_I . The tensor-to-scalar ratio and tensor tilt are

$$r(k) = \frac{\Delta_T^2(k)}{\Delta_\zeta^2(k)}, \quad n_T = \frac{d \ln \Delta_T^2}{d \ln k}. \quad (22)$$

These quantities are especially useful compactification diagnostics because internal geometry can modify the scalar and tensor blocks differently. A shift in A_s can often be absorbed into a change of the inflationary potential. A scale-dependent r , extra tensor polarizations, or a violation of $r = -8n_T$ is harder to imitate.

For rapid screening we use

$$A_s \simeq 2.1 \times 10^{-9}, \quad 0.955 < n_s < 0.980, \quad r_{0.05} < 0.036, \quad (23)$$

consistent with Planck and BICEP/Keck analyses [5–7]. These are not a substitute for a likelihood analysis; they are pass/fail benchmarks.

A. Recovery of the standard result

The reduced quadratic action for a single adiabatic mode is

$$S_\zeta^{(2)} = \frac{1}{2} \int d\eta \frac{d^3 k}{(2\pi)^3} z^2 (|\zeta'_k|^2 - c_s^2 k^2 |\zeta_k|^2), \quad (24)$$

where $z^2 = 2a^2 \epsilon \bar{M}_P^2 / c_s^2$. For canonical slow roll, $c_s = 1$. In approximate de Sitter space, $a = -1/(H\eta)$, and the regular solution may be normalized as

$$u_k(\eta) = (1 - ik\eta) e^{ik\eta}. \quad (25)$$

The boundary kernel is $K_\zeta = z^2 u'_k / u_k$, giving

$$\operatorname{Im} K_\zeta(k, \eta) = \frac{2\epsilon \bar{M}_P^2}{H^2} \frac{k^3}{1 + k^2 \eta^2}. \quad (26)$$

At late times $|k\eta| \ll 1$,

$$\operatorname{Im} K_\zeta \longrightarrow \frac{2\epsilon \bar{M}_P^2}{H^2} k^3. \quad (27)$$

Equations (9) and (17) then give

$$\Delta_\zeta^2 = \frac{\hbar H_*^2}{8\pi^2 \epsilon_* \bar{M}_P^2}, \quad (28)$$

while the tensor block gives

$$\Delta_T^2 = \frac{2\hbar H_*^2}{\pi^2 \bar{M}_P^2}, \quad r = 16\epsilon_*. \quad (29)$$

The method therefore reproduces the textbook answer. Its value is not a different prediction in the decoupling limit, but a cleaner route for deciding when a compactification does not lie in that limit.

IV. HOW COMPACTIFICATION DATA ENTER

Consider a $D = 4 + d$ dimensional metric

$$ds_D^2 = e^{2A(\phi)} g_{\mu\nu}(x) dx^\mu dx^\nu + G_{mn}(y; \phi) dy^m dy^n, \quad (30)$$

where $\phi^a(x)$ are breathing, shape, and flux-supported moduli. In the four-dimensional Einstein frame,

$$S_4 = \int d^4x \sqrt{-g} \left[\frac{\bar{M}_P^2}{2} R - \frac{1}{2} G_{ab}(\phi) \partial_\mu \phi^a \partial^\mu \phi^b - V_{\text{eff}}(\phi) \right] + S_{\text{light}}. \quad (31)$$

Fluctuations on the internal manifold are expanded in eigenfunctions of the appropriate scalar, vector, spinor, and Lichnerowicz operators. For a scalar harmonic,

$$-\nabla_{X_d}^2 Y_I = \lambda_I Y_I, \quad \chi(x, y) = \sum_I \chi_I(x) Y_I(y). \quad (32)$$

The four-dimensional masses contain λ_I plus curvature, flux, and interaction contributions. The internal geometry therefore enters through the spectral data (λ_I, d_I) and overlap integrals that determine mixing.

Separate observable variables $L = (\zeta, h_+, h_\times)$ from internal variables $H = (s^\alpha, \chi^r)$. The quadratic boundary kernel is

$$K = \begin{pmatrix} K_{LL} & K_{LH} \\ K_{HL} & K_{HH} \end{pmatrix}. \quad (33)$$

Integrating out unobserved internal variables gives

$$K_{L,\text{eff}} = K_{LL} - K_{LH} K_{HH}^{-1} K_{HL}. \quad (34)$$

This Schur complement is the compactification correction. It contains changes in normalization, scale-dependent mixing, resonant structure, and the onset of instability when an eigenvalue of K_{HH} approaches zero.

When an internal mode has mass $M \gg H$ and varies adiabatically,

$$\delta K_L \sim -\frac{K_{LH} K_{HL}}{M^2} = \mathcal{O}\left(\frac{H^2}{M^2}\right) K_{LL}, \quad (35)$$

up to couplings and slow-roll factors. Distinct manifolds then become observationally degenerate. If $M \sim H$, corrections can be large and may appear in power-spectrum features or higher-point functions [8].

For numerical work one may evolve the kernel directly. For a canonically normalized vector v obeying

$$v'' + \Omega^2(k, \eta) v = 0, \quad (36)$$

let F be a fundamental matrix of contour-selected solutions. The kernel is

$$K = F' F^{-1}, \quad (37)$$

up to the kinetic prefactor, and satisfies the Riccati equation

$$K' + K^2 + \Omega^2 = 0. \quad (38)$$

This avoids separately normalizing every mixed mode and is convenient for scanning flux sectors or manifolds.

V. WORKED EXAMPLE: $M_4 \times S^2 \times \mathbb{CP}^2$

A. Background and flux branches

$$ds_{10}^2 = e^{2A} g_{\mu\nu} dx^\mu dx^\nu + a_1^2 d\Omega_2^2 + a_2^2 ds_{\mathbb{CP}^2}^2, \quad (39)$$

where the reference Fubini–Study metric is normalized once and all powers of a_2 are explicit. The internal volume scales as

$$V_6 \propto a_1^2 a_2^4. \quad (40)$$

The two logarithmic radii may be written

$$u = \ln(a_1/a_{1,0}), \quad v = \ln(a_2/a_{2,0}). \quad (41)$$

In an unconstrained two-radion reduction these fields have the Einstein-frame kinetic metric derived in Appendix B. If the UECKK unimodular condition is imposed as exact preservation of the internal volume, the combination

$$\sigma = 2u + 4v = \ln(V_6/V_{6,0}) \quad (42)$$

is fixed and only the shape deformation orthogonal to volume remains dynamical. The physical shape coordinate may then be taken as $\chi = u - v$, with the projected metric given in Eq. (B14).

A useful flux-and-curvature truncation is

$$U(a_1, a_2) = -A \left[\frac{2}{a_1^4 a_2^4} + \frac{24}{a_1^2 a_2^6} \right] + B \left[\frac{m^2}{2a_1^6 a_2^4} + \frac{qn^2}{a_1^2 a_2^8} \right] - C \frac{N_{\text{fam}}^2}{a_1^2 a_2^4}. \quad (43)$$

The first line is the internal-curvature contribution in the chosen reduction convention, the second contains quantized flux, and the final term is a volume-dependent attractive contribution. The coefficients A, B, C, q are model parameters until derived from the microscopic action. The family index is

$$N_{\text{fam}} = m \frac{n^2 - 1}{8}. \quad (44)$$

Both (3, 3) and (1, 5) give $N_{\text{fam}} = 3$, but their flux terms are

$$U_{\text{flux}}^{(3,3)} = B \left[\frac{9}{2a_1^6 a_2^4} + \frac{9q}{a_1^2 a_2^8} \right], \quad (45)$$

$$U_{\text{flux}}^{(1,5)} = B \left[\frac{1}{2a_1^6 a_2^4} + \frac{25q}{a_1^2 a_2^8} \right]. \quad (46)$$

They therefore have different minima, modulus Hessians, and overlap tensors even though their net chiral family number is identical.

The first background test is algebraic:

$$\partial_{a_i} U = 0, \quad \text{eigenvalues } [G^{-1} \nabla \nabla U] > 0. \quad (47)$$

The first condition locates a compactification saddle; the second tests local stability in the homogeneous truncation. A negative eigenvalue rejects the branch before any CMB calculation.

B. Internal spectra

For a sphere of radius a_1 , scalar harmonics obey

$$\lambda_\ell^{S^2} = \frac{\ell(\ell+1)}{a_1^2}, \quad d_\ell^{S^2} = 2\ell+1. \quad (48)$$

For the standard Fubini–Study normalization on $\mathbb{CP}^2$,

$$\lambda_p^{\mathbb{CP}^2} = \frac{4p(p+2)}{a_2^2}, \quad d_p^{\mathbb{CP}^2} = (p+1)^3, \quad (49)$$

with a rescaling if a different curvature convention is used. Product harmonics have

$$\lambda_{\ell p} = \lambda_\ell^{S^2} + \lambda_p^{\mathbb{CP}^2}. \quad (50)$$

Flux shifts the spectra of charged fields and changes the internal wave functions. That shift is essential for distinguishing (3,3) from (1,5): the branches have the same index but not the same mode profiles or mixing integrals.

C. Quadratic kernel

Retain the adiabatic mode, the two homogeneous moduli, and a finite set of KK modes,

$$q = (\zeta, u, v, h_+, h_-, \chi_{\ell p}^{(r)}). \quad (51)$$

After solving the four-dimensional lapse and shift constraints, the scalar block has the schematic action

$$S_S^{(2)} = \frac{1}{2} \int d\eta \frac{d^3 k}{(2\pi)^3} [Q'^T Q' - Q^T \Omega_S^2(k, \eta) Q], \quad (52)$$

where Q is the canonically normalized scalar vector. The matrix Ω_S^2 contains the field-space curvature, the canonically normalized Hessian of U , the KK eigenvalues, and mixing terms proportional to the turning rate of the background trajectory. The contour-selected fundamental matrix F_S gives

$$K_S = F_S' F_S^{-1}. \quad (53)$$

The effective curvature kernel follows from Eq. (34). The tensor block is computed independently, including any mixing of the four-dimensional graviton with internal transverse-traceless metric modes.

For a rapid branch comparison, define

$$\mathcal{D}_{(m,n)}(k) = \{\text{eig } K_I, A_s, n_s, \alpha_s, r, n_T, \beta_{\text{iso}}\}_{(m,n)}. \quad (54)$$

The two branches are distinguishable only if

$$\mathcal{D}_{(3,3)}(k) \neq \mathcal{D}_{(1,5)}(k) \quad (55)$$

by more than observational and theoretical uncertainties. If both modulus masses satisfy $M_u, M_v \gg H$ and all charged KK modes are heavier still, both branches reduce to the universal single-field kernel. The method then returns a useful null result: the CMB cannot discriminate the flux distribution at quadratic order.

D. What can make the branches differ

There are four leading possibilities. First, one branch may be unstable. Second, one may have a modulus with $M \sim H$, producing isocurvature or a scale-dependent Schur correction. Third, the time dependence of $V_6 \propto a_1^2 a_2^4$ may modify the tensor normalization through an evolving effective Planck mass. Fourth, flux-dependent KK levels may lie near H and leave oscillatory or non-Gaussian signatures. The scalar power-law spectrum alone is unlikely to identify the branch if both are deep in the decoupling regime.

VI. $M_4 \times S^3 \times S^3$ AS A CONTROL

Consider

$$ds_{10}^2 = e^{2A} g_{\mu\nu} dx^\mu dx^\nu + b_1^2 d\Omega_{3,1}^2 + b_2^2 d\Omega_{3,2}^2. \quad (56)$$

This product has the same number of internal dimensions as $S^2 \times \mathbb{CP}^2$ but different curvature and spectral data. The internal volume scales as $b_1^3 b_2^3$. Two logarithmic radii again give a volume and a shape mode.

For a unit three-sphere the scalar harmonics satisfy

$$\lambda_\ell^{S^3} = \frac{\ell(\ell+2)}{b^2}, \quad d_\ell^{S^3} = (\ell+1)^2. \quad (57)$$

Product eigenvalues are

$$\lambda_{\ell_1 \ell_2} = \frac{\ell_1(\ell_1+2)}{b_1^2} + \frac{\ell_2(\ell_2+2)}{b_2^2}. \quad (58)$$

A stabilizing model may include curvature, a cosmological term, and quantized three-form flux on each factor. Its Einstein-frame potential can be written generally as

$$U_{33}(b_1, b_2) = U_{\text{curv}} + U_{\text{flux}} + U_{\text{other}}, \quad (59)$$

with each term carrying a fixed power of b_1 and b_2 determined by dimensional reduction. The kernel algorithm is unchanged: locate a stable saddle, canonically normalize the two moduli, construct Ω^2 , solve the regular matrix mode problem, and take the Schur complement.

This control example makes the main methodological point. The observable formulas do not need to be reinvented. What changes is the internal input:

$$\{G_{ab}, U, \lambda_I, d_I, \text{overlaps}\}_{S^2 \times \mathbb{CP}^2} \longrightarrow \{G_{ab}, U, \lambda_I, d_I, \text{overlaps}\}_{S^3 \times S^3}. \quad (60)$$

The same code then produces $K_{\zeta, \text{eff}}$ and $K_{T, \text{eff}}$. A proposed universe is rejected if either kernel fails the criteria below. If both manifolds pass and their internal modes decouple, their CMB two-point spectra are not a useful discriminant; one must turn to particle content, non-Gaussianity, or other observables.

TABLE I. Dimensionless benchmark minima and eigenvalues of $G^{-1}\nabla\nabla U$. These values demonstrate the pipeline and are not microscopic predictions.

Model	a_1 or b_1	a_2 or b_2	m_-^2	m_+^2
(3, 3) on $S^2 \times \mathbb{CP}^2$	0.929	0.725	122.6	255.6
(1, 5) on $S^2 \times \mathbb{CP}^2$	0.535	1.151	34.1	72.1
$S^3 \times S^3$ control	1.033	1.033	12.1	30.3

VII. NUMERICAL DEMONSTRATION OF THE SCREENING PIPELINE

The preceding sections define a method, but a useful screening paper should also show that the chain from compactification data to an observational curve is operational. We therefore give a transparent benchmark calculation. The chosen dimensionless coefficients produce stable saddles and expose each computational step. They are not yet microscopic predictions of either compactification; a genuine verdict on a manifold requires coefficients and mixing matrices derived from its higher-dimensional action.

For a product of two Einstein spaces of dimensions d_1 and d_2 , with logarithmic radii $\beta_i = \ln a_i$, the unconstrained Einstein-frame kinetic matrix in the homogeneous truncation is

$$G_{ij} = \bar{M}_P^2 \left(d_i \delta_{ij} + \frac{1}{2} d_i d_j \right). \quad (61)$$

Thus, in units $\bar{M}_P = 1$,

$$G_{S^2 \times \mathbb{CP}^2} = \begin{pmatrix} 4 & 4 \\ 4 & 12 \end{pmatrix}, \quad G_{S^3 \times S^3} = \begin{pmatrix} 15/2 & 9/2 \\ 9/2 & 15/2 \end{pmatrix}. \quad (62)$$

For the illustrative benchmark in this section we keep the two-radion matrix so that the screening algorithm can be displayed in its full matrix form. In a strictly unimodular implementation of the $S^2 \times \mathbb{CP}^2$ model, the volume direction is projected out as shown in Appendix B; then the surviving homogeneous modulus has $G_{\chi\chi} = 4\bar{M}_P^2/3$. For $S^2 \times \mathbb{CP}^2$ we set $A = B = q = 1$ and $C = 0.05$ in Eq. (43). For the control model we use

$$U_{33} = -6(b_1^{-5}b_2^{-3} + b_1^{-3}b_2^{-5}) + \frac{9}{2}(b_1^{-9}b_2^{-3} + b_1^{-3}b_2^{-9}) + 0.2b_1^{-3}b_2^{-3}. \quad (63)$$

The extrema are found in logarithmic variables and the canonical mass matrix is $G^{-1}\nabla\nabla U$. Table I lists the results. All three examples possess a stable homogeneous saddle in this benchmark, but their radii and canonical mass eigenvalues differ substantially.

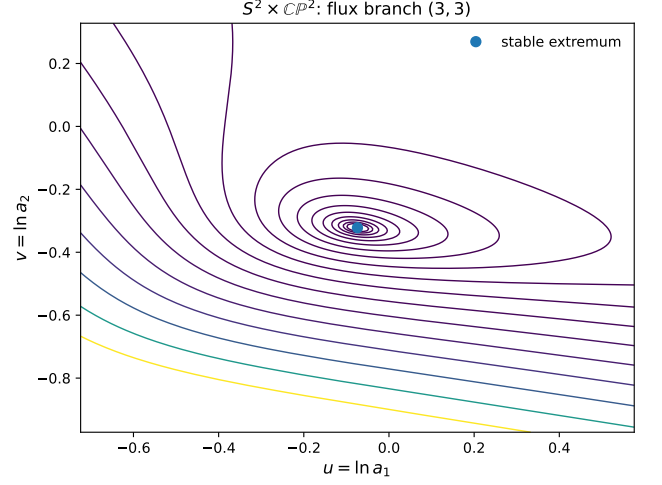


FIG. 1. Benchmark potential for the (3, 3) branch on $S^2 \times \mathbb{CP}^2$. The dot marks the stable extremum.

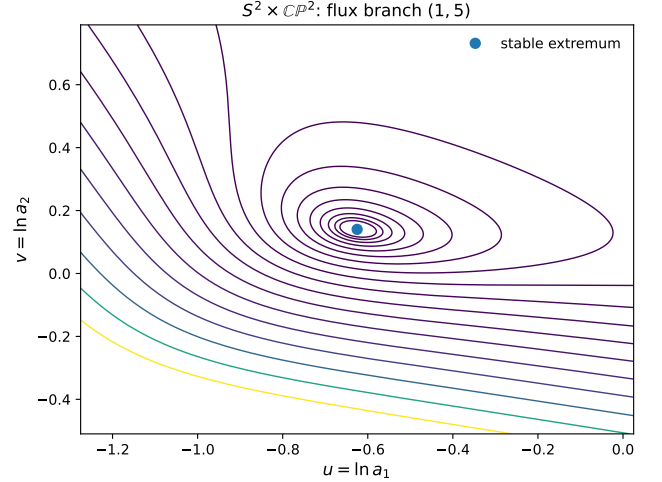


FIG. 2. Benchmark potential for the (1, 5) branch. Although it has the same family index as (3, 3), its flux distribution moves the minimum and changes the local Hessian.

A. Observed amplitude and the tensor constraint

Combining $A_s = H_*^2/(8\pi^2\epsilon_*\bar{M}_P^2)$ with $r = 16\epsilon_*$ and $M_P = \sqrt{8\pi}\bar{M}_P$ gives

$$\frac{H_*}{M_P} = \frac{1}{4}\sqrt{\pi A_s r}. \quad (64)$$

Figure 3 uses $A_s = 2.10 \times 10^{-9}$ and the BICEP/Keck limit $r_{0.05} < 0.036$ [6, 7]. In the canonical single-field limit the bound corresponds to $H_*/M_P < 3.9 \times 10^{-6}$. A proposed compactification supplies the background value of H_* and the tensor kernel supplies r ; their point must lie on this curve.

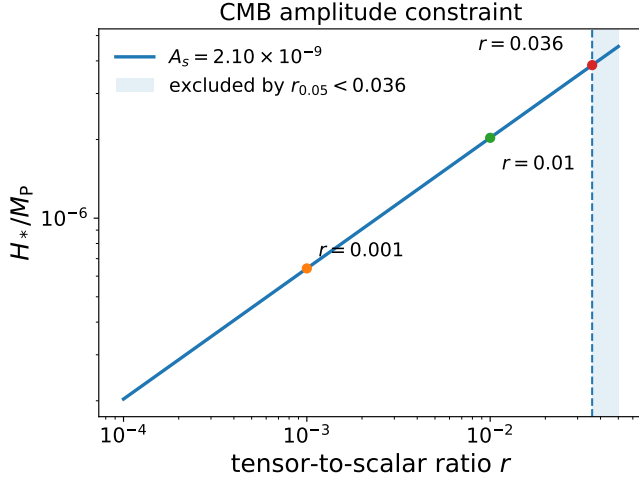


FIG. 3. The Hubble scale required by the measured scalar amplitude as a function of r . The shaded region is excluded by the current BICEP/Keck bound.

B. A one-mode Schur-complement visualization

To display the last algebraic step, retain one internal scalar with dimensionless mass $\mu = M/H$ and approximate the imaginary kernel near the pivot by

$$\frac{\text{Im } K_{\zeta, \text{eff}}}{\text{Im } K_{\zeta, 0}} = 1 - \frac{\gamma^2}{\mu^2 + (k/k_*)^2}. \quad (65)$$

Since the power is inversely proportional to $\text{Im } K$, the response is

$$\frac{\Delta_{\zeta}^2}{\Delta_{\zeta, 0}^2} = \left[1 - \frac{\gamma^2}{\mu^2 + (k/k_*)^2} \right]^{-1}. \quad (66)$$

For Fig. 4 we set the common illustrative values $H^2 = 10$ and $\gamma = 0.4$, and insert the lighter canonical mass from Table I. The calculation is deliberately simple enough to reproduce by inspection. A full model replaces Eq. (65) by the Riccati evolution of the constrained matrix kernel.

C. Direct comparison with the measured scalar spectrum

Figure 5 carries the benchmark through to an observational screen. The band is the pointwise envelope generated by the Planck 2018 constraints $n_s = 0.9649 \pm 0.0042$ and $\ln(10^{10} A_s) = 3.044 \pm 0.014$ [4, 5]. Each benchmark curve is normalized to the measured amplitude at $k_* = 0.05 \text{ Mpc}^{-1}$; the test therefore probes the additional scale dependence produced by the internal mode rather than refitting the amplitude.

With the illustrative choices above, the (3,3) curve remains inside this one-standard-deviation parameter envelope over about 94% of the plotted logarithmic k grid,

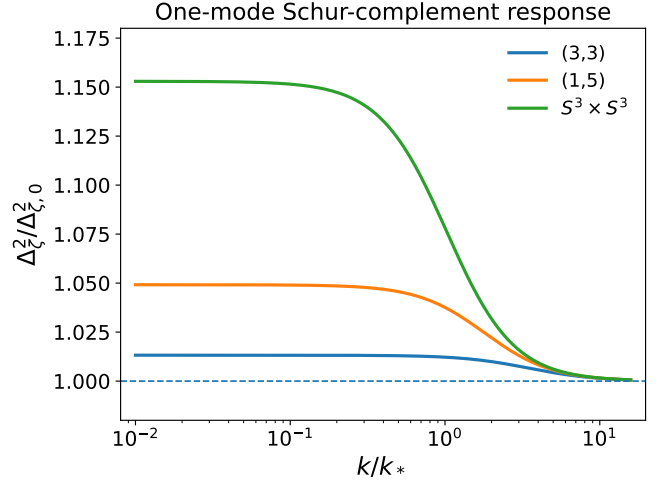


FIG. 4. Illustrative scalar-power response from one internal mode. The different mass eigenvalues generate visibly different scale dependences.

the (1,5) curve over about 28%, and the $S^3 \times S^3$ control over about 5%. These percentages are not likelihoods and must not be interpreted as observational confidence levels. They demonstrate that the kernel method can return a positive or negative screening verdict once the compactification masses and mixings are supplied. In this benchmark the (3,3) branch is close to the observed power-law shape, whereas the lighter internal modes of the other examples generate excessive low- k curvature.

The final qualification is essential. The present plot proves that the procedure works and shows which derived quantities control the answer. It does not yet provide positive evidence for the manifold, because changing the benchmark mixing changes the curves. A fourth companion paper on $M_4 \times S^2 \times \mathbb{CP}^2$ would become possible when G_{ab} , the full quadratic mixing matrix, and the inflationary trajectory are derived from the same microscopic model used in the family, torsion, and stability papers.

VIII. A PRACTICAL CMB ACCEPTABILITY TEST

The boundary-kernel method is intended as a staged filter. It does not eliminate the final Boltzmann calculation for a successful model. It reduces the number of models that reach that stage.

A. Stage 1: background viability

For each manifold and flux sector:

1. Reduce to the four-dimensional Einstein frame and compute $G_{ab}(\phi)$ and $V_{\text{eff}}(\phi)$.

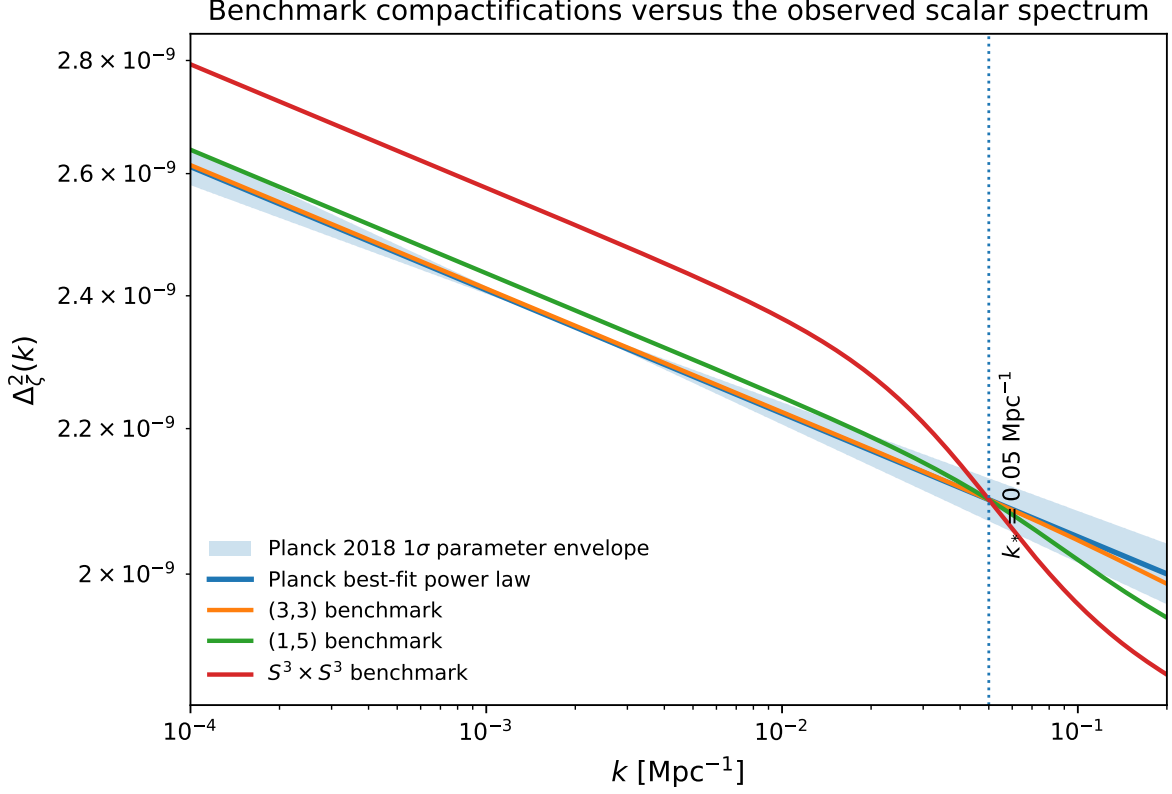


FIG. 5. Benchmark primordial spectra compared with the Planck 2018 best-fit power law and its one-standard-deviation parameter envelope. This is an observational comparison, but not yet a manifold prediction: the benchmark mixing γ and the overall potential normalization have not been derived from either higher-dimensional action. A genuine confirmation or exclusion of $M_4 \times S^2 \times \mathbb{CP}^2$ requires those microscopic inputs.

2. Find a background solution with stabilized internal dimensions or a controlled slow trajectory.
3. Require positive kinetic eigenvalues and no tachyonic instability faster than the cosmological time scale.
4. Verify that the KK gap remains above the cutoff used in the truncation.

A model failing this stage is not a candidate four-dimensional universe.

B. Stage 2: kernel viability

Construct the constrained scalar and tensor kernels on a grid of wavenumbers covering the observable CMB interval. Require

$$\text{eig } K_I(k) > 0 \quad (67)$$

for every retained physical mode and every relevant k . A zero or negative eigenvalue identifies a nonnormalizable direction, a bad contour choice, or a physical instability. Next integrate out modes not retained as observables and verify that the Schur complement is numerically stable.

C. Stage 3: primordial observables

From the effective kernels calculate

$$A_s, \quad n_s, \quad \alpha_s, \quad r, \quad n_T, \quad \beta_{\text{iso}}, \quad (68)$$

where β_{iso} is the fractional isocurvature power. Apply the broad windows in Eq. (23) and an appropriately stringent isocurvature bound. A model passing these cuts is worth a full likelihood analysis.

D. Stage 4: distinguishability

For two candidate manifolds $\mathcal{M}_1$ and $\mathcal{M}_2$, define the fractional kernel separation

$$\Xi_X(k) = \frac{\left| \text{Im } K_X^{(1)}(k) - \text{Im } K_X^{(2)}(k) \right|}{\frac{1}{2} \left[\text{Im } K_X^{(1)}(k) + \text{Im } K_X^{(2)}(k) \right]}, \quad (69)$$

for $X = \zeta, T$. If Ξ_X is smaller than the combined observational and theoretical uncertainty throughout the observable window, the two compactifications are CMB-degenerate at the level of two-point functions. This is not

a failure of the method. It is the correct statement that the internal geometry has decoupled.

IX. DISCUSSION

A. What is new and what is not

The identification of the Gaussian wave-functional width with the primordial covariance is not new. Neither is the standard inflationary result recovered in Sec. III. The proposed contribution is a calculational organization: the constrained Lorentzian boundary kernel is used as a common comparison object for candidate higher-dimensional universes. The same matrix formalism handles moduli, KK towers, scalar-tensor mixing, and boundary relocation.

This organization is useful only if it saves work or exposes a failure earlier than the conventional pipeline. The main savings are threefold. First, positivity of the imaginary kernel tests stability and normalizability before a full transfer-function calculation. Second, the Schur complement states immediately whether heavy internal modes decouple. Third, evolving a matrix Riccati equation can be simpler than separately normalizing a large set of coupled mode functions.

B. Can the CMB determine the manifold?

Generally no. Distinct manifolds can produce the same four-dimensional light theory, and even spectral data need not uniquely determine a geometry. The realistic outcomes are exclusion of a branch, distinction between broad classes, or constraints on modulus masses and couplings. The scalar two-point spectrum is already close to a featureless power law, so viable manifolds must reproduce nearly the same scalar kernel. Tensor modes, isocurvature, non-Gaussianity, and flux-dependent massive exchanges are more promising discriminants.

For the $(3, 3)$ and $(1, 5)$ branches, the family index alone cannot distinguish them. The difference lies in the flux distribution, the internal wave-function profiles, and the resulting modulus and KK kernels. If both branches have all internal masses far above H , the CMB cannot tell them apart at quadratic order. If one branch contains a mode near the Hubble scale, the method identifies the corresponding correction and indicates which observable should be computed next.

C. Relation to an early high-curvature boundary

The kernel formalism does not require the initial boundary to lie at a particular curvature. Moving an intermediate boundary merely redistributes the description between an assigned boundary state and subsequent propagation, provided the same contour and dynamics are used. A

specific quantum-boundary proposal may supply a physical earliest surface, but it is not needed for the kernel method itself.

Similarly, a short decelerating period can lower a high initial Hubble rate before inflation, but that standard Friedmann scaling is not the source of the observed fractional perturbation amplitude. The amplitude is fixed by the kernel when observable modes exit the inflationary horizon, through the combination H_*^2/ϵ_* . The present formalism cleanly separates that universal statement from compactification-specific dynamics.

D. Limitations

The method is semiclassical and quadratic. It assumes a controllable saddle, a well-defined contour, and a truncation whose omitted modes do not invalidate the Schur complement. A candidate passing the two-point test may still fail through non-Gaussianity, reheating, defects, late-time moduli, or particle-physics constraints. Near the compactification or quantum-gravity scale, higher-derivative operators and the full KK tower may be important. These are not technical footnotes: they define the domain in which the screening calculation is trustworthy.

X. CONCLUSION

A higher-dimensional cosmological model can be screened against the CMB through its constrained Lorentzian boundary kernel. The imaginary part of that kernel is the inverse covariance of the physical boundary variables. Scalar and tensor spectra are therefore read from the same object that tests stability and encodes mixing with internal modes.

The method reproduces the standard single-field inflationary spectrum when moduli and KK modes decouple. Its practical value appears away from that limit. For $M_4 \times S^2 \times \mathbb{CP}^2$, the $(3, 3)$ and $(1, 5)$ flux branches have the same chiral family index but different flux support, modulus Hessians, charged spectra, and overlap tensors. These differences enter the effective scalar and tensor kernels through a Schur complement. For $M_4 \times S^3 \times S^3$, the same algorithm applies with different curvature and harmonic data. No new cosmological formalism is required when the manifold is changed; only the internal inputs change.

The method is best viewed as an acceptability filter. It can reject unstable compactifications, identify light internal modes, quantify decoupling, and determine whether a full CMB likelihood calculation is warranted. It cannot generally infer a unique manifold from a nearly featureless scalar spectrum. A successful discrimination will most likely involve tensor modes, isocurvature, non-Gaussianity, or a characteristic departure from the four-dimensional consistency relations.

TABLE II. Minimal inputs and outputs of the boundary-kernel screening test.

Stage	Required input	Immediate output
Background	G_{ab} , V_{eff} , flux integers	stable saddle, $H(t)$, modulus masses
Internal spectrum	Laplace/Lichnerowicz eigenvalues and overlaps	KK masses, degeneracies, mixing matrices
Quadratic dynamics	constrained scalar and tensor actions	matrix frequencies Ω_S^2 , Ω_T^2
Lorentzian state	regularity or $i\epsilon$ contour	positive imaginary boundary kernel
CMB screen	$K_{\zeta, \text{eff}}$, $K_{T, \text{eff}}$	$A_s, n_s, \alpha_s, r, n_T, \beta_{\text{iso}}$
Discrimination	two or more candidate kernels	exclusion, distinguishability, or decoupling bound

Equivalent formulations of a quantum theory need not be equally transparent for every problem. Canonical and in-in methods are particularly effective for perturbative evolution on a chosen background. The boundary-wave-functional representation instead emphasizes the global state, its contour and boundary data, and the possibility of combining multiple sectors or saddles before observables are extracted. At quadratic order these descriptions agree. Beyond that order, the same organization may help identify contributions that are easy to suppress when the background, state, topology, or compactification sector is fixed at the outset. Establishing whether any such contribution survives as a gauge-invariant observable is a separate problem, but the kernel hierarchy provides a natural framework in which to ask it.

Appendix A: Gaussian integration and the Schur complement

Let a Gaussian state depend on observed variables x and hidden variables y ,

$$\Psi(x, y) \propto \exp \left[\frac{i}{2\hbar} \begin{pmatrix} x & y \end{pmatrix} \begin{pmatrix} A & B \\ B^T & C \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} \right]. \quad (\text{A1})$$

Integrating over y gives

$$\Psi_{\text{eff}}(x) \propto (\det C)^{-1/2} \exp \left[\frac{i}{2\hbar} x^T (A - BC^{-1}B^T) x \right], \quad (\text{A2})$$

with the contour chosen so that the Gaussian converges. This proves Eq. (34). For a field theory the matrices carry momentum labels and functional determinants replace ordinary determinants.

Appendix B: Radion metric for $S^2 \times \mathbb{CP}^2$ and the unimodular projection

This appendix derives the homogeneous radion kinetic metric used in the worked example and then projects it to the volume-preserving subspace appropriate to a strict unimodular interpretation. The distinction is important. The unconstrained two-radion metric is useful for displaying the matrix screening method, while the exact UECKK

model should use the projected metric if unimodularity fixes the internal volume.

Consider the $D = 4 + d$ dimensional metric

$$ds_D^2 = e^{2A(x)} g_{\mu\nu}(x) dx^\mu dx^\nu + \sum_i e^{2\beta_i(x)} ds_i^2, \quad (\text{B1})$$

where the i th internal factor has dimension d_i and fixed unit-volume metric ds_i^2 . Define

$$\Sigma = \sum_i d_i \beta_i = \ln(V_d/V_{d,0}). \quad (\text{B2})$$

The four-dimensional Einstein frame is obtained by choosing

$$A = -\frac{\Sigma}{D_4 - 2} = -\frac{\Sigma}{2}, \quad (\text{B3})$$

for $D_4 = 4$. Reducing the Einstein–Hilbert term, dropping total derivatives, and integrating over the reference internal space gives

$$S_{\text{EH}} = \frac{\bar{M}_{\text{P}}^2}{2} \int d^4x \sqrt{-g} [R - \mathcal{G}_{ij} \partial_\mu \beta_i \partial^\mu \beta_j + \dots], \quad (\text{B4})$$

with

$$\mathcal{G}_{ij} = d_i \delta_{ij} + \frac{d_i d_j}{D_4 - 2} = d_i \delta_{ij} + \frac{1}{2} d_i d_j. \quad (\text{B5})$$

Equivalently, in the matter-field convention

$$S_{\text{kin}} = -\frac{1}{2} \int d^4x \sqrt{-g} G_{ij} \partial_\mu \beta_i \partial^\mu \beta_j, \quad (\text{B6})$$

one has $G_{ij} = \bar{M}_{\text{P}}^2 \mathcal{G}_{ij}$.

For $S^2 \times \mathbb{CP}^2$, $d_1 = 2$, $d_2 = 4$, and $(\beta_1, \beta_2) = (u, v)$. Therefore

$$G_{uv} = \begin{pmatrix} 4 & 4 \\ 4 & 12 \end{pmatrix}, \quad G_{uv} = \bar{M}_{\text{P}}^2 \begin{pmatrix} 4 & 4 \\ 4 & 12 \end{pmatrix}. \quad (\text{B7})$$

The corresponding line element on the two-radion field space is

$$ds_{\text{rad}}^2 = 4du^2 + 8dudv + 12dv^2. \quad (\text{B8})$$

It is useful to change variables to total internal volume and relative shape,

$$\sigma = 2u + 4v, \quad \chi = u - v. \quad (\text{B9})$$

Solving for u and v ,

$$u = \frac{\sigma}{6} + \frac{2\chi}{3}, \quad v = \frac{\sigma}{6} - \frac{\chi}{3}, \quad (\text{B10})$$

and substituting gives the diagonal form

$$ds_{\text{rad}}^2 = \frac{2}{3}d\sigma^2 + \frac{4}{3}d\chi^2. \quad (\text{B11})$$

Thus the unconstrained canonical fields are

$$\Phi_\sigma = \bar{M}_P \sqrt{\frac{2}{3}} \sigma, \quad \Phi_\chi = \bar{M}_P \sqrt{\frac{4}{3}} \chi. \quad (\text{B12})$$

If unimodularity is imposed as exact preservation of internal volume under allowed deformations, then

$$\delta\sigma = 2\delta u + 4\delta v = 0. \quad (\text{B13})$$

The volume mode σ is not a propagating radion. The induced field-space metric on the allowed one-dimensional shape subspace is therefore

$$ds_{\text{uni}}^2 = \frac{4}{3}d\chi^2, \quad \boxed{G_{\chi\chi}^{\text{uni}} = \frac{4}{3}\bar{M}_P^2}. \quad (\text{B14})$$

Equivalently, using $v = v_0 - u/2$ along a fixed-volume curve gives

$$ds_{\text{uni}}^2 = 3du^2, \quad \Phi_u = \sqrt{3}\bar{M}_P u, \quad (\text{B15})$$

which is the same result because $d\chi = (3/2)du$.

For a potential $U(u, v)$, the strict unimodular shape potential is the pullback

$$U_{\text{uni}}(\chi) = U\left(\frac{\sigma_0}{6} + \frac{2\chi}{3}, \frac{\sigma_0}{6} - \frac{\chi}{3}\right), \quad (\text{B16})$$

where σ_0 is fixed by the selected internal volume. At a stationary point the physical shape mass is

$$m_\chi^2 = \frac{1}{G_{\chi\chi}^{\text{uni}}} \frac{\partial^2 U_{\text{uni}}}{\partial \chi^2} = \frac{3}{4\bar{M}_P^2} \frac{\partial^2 U_{\text{uni}}}{\partial \chi^2}. \quad (\text{B17})$$

In two-coordinate notation this same projection is

$$m_\chi^2 = \frac{p^I H_{IJ} p^J}{\bar{M}_P^2 p^I G_{IJ} p^J}, \quad p^I = \left(\frac{2}{3}, -\frac{1}{3}\right), \quad (\text{B18})$$

where $H_{IJ} = \partial_I \partial_J U$ at the extremum and $p^I \partial_I = \partial/\partial \chi$ at fixed σ . This is the quantity that should be compared with H_*^2 in a strict volume-preserving UECKK CMB screen.

Appendix C: Suggested numerical workflow

A minimal implementation for each k is:

1. evaluate the background and canonical kinetic matrix;
2. assemble $\Omega^2(k, \eta)$ for all retained modes;
3. initialize the regular positive-frequency fundamental matrix where k/a exceeds every background scale in the retained system;
4. evolve either F or the Riccati matrix K ;
5. project onto the physical boundary basis;
6. check positivity of K_I ;
7. integrate out hidden modes by Eq. (34);
8. calculate the observables in Sec. III.

The procedure is repeated over a logarithmic k grid and over the compactification parameters or flux integers to be screened.

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